Twelve integers at degree 2048

Round thirteen research contribution

24 October 2026

1 Twelve integers at degree 2048

A late supplied interpolation theorem permits a further synthesis. The result is a literal polynomial in twelve unrestricted integer variables, of exact degree 2048, evaluated in 233 live binary operations (148 multiplications, 53 additions and 32 subtractions). It improves the supplied twelve-variable 4832/248 degree–cost point. The supplied 16000/223 point remains a cheaper, higher-degree alternative. No fewest-variable or global optimality claim is made.

The new changes are a coprime homogeneous quintic carrier, a guard Pell witness divisible by three, fixed prime separation in place of large size weights, and a doubled auxiliary-root center. The full-residue interpolation theorem and the direct normalized signed-kernel calibration are imported from `ConZFC_Twelve_Integer_Interpolation_2026-10-24`. Its exact proofs, provenance and notices are retained. In particular, this is not a literal restoration of the earlier odd-multiplier source chart, nor a new proof-assistant certificate.

1.1 Complete polynomial specification

The twelve coordinates are

$$j, C, z, h, r, v, w, T, c, n, t, s_0 \in \mathbb{Z}.$$

Use the same complete six fixed source-34/source-71 numerals as before. Define

$$\begin{aligned}
J &= 6j + 1, & q &= \beta J + 1, & W &= 12w + 4,\\
X &= qW, & Y &= qv, & E &= XY,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
F &= (K_0 + W)C - z(q - 1),\\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J,\\
k &= R + 1 + hE, & \eta &= c - Yk, & \zeta &= k - \eta.
\end{aligned}$$

Put

$$
\begin{aligned}
d_g &= (A_{\mathrm{cone}}F)(B_{\mathrm{cone}}C) - (q - A_{\mathrm{cone}}F - B_{\mathrm{cone}}C)^2, \\
P &= d_g\eta\zeta - (t^2 - d_g - \eta\zeta + 1)^2, \\
s &= 3s_0, & D_g &= (3P-4)s^2 + 1, & I &= src^2, \\
D_f &= 1 + \Delta I^2, & Q &= \Delta^2 I^2, & \delta &= Q - 1, \\
V &= D_f(c - R + 4IT) - c, & C_y &= QV^2 - 1, & D_a &= \delta C_y, \\
\ell &= EYk, & D_\tau &= 1 + \ell(\ell + k), & D_m &= 1 + \Delta c^2, \\
b_0 &= X + ac, & M &= 2q^2, & L &= MH, \\
e &= v - q^2, & \widetilde{b} &= Mb_0 + He, & u &= LT - \widetilde{b}, \\
\widetilde{D} &= M^2 D_m.
\end{aligned}
$$

For the fixed intersective quintic

$$\pi(x) = (x^2 + 31)(x^3 + x + 1), \qquad \Phi_L(x) = L^5\pi(x/L),$$

the following integer polynomials are its exact quadratic remainder:

$$
\begin{aligned}
A &= u^5 + 10u^3\widetilde{D} + 5u\widetilde{D}^2 + 32L^2(u^3 + 3u\widetilde{D}) \\
&\quad + L^3(u^2 + \widetilde{D}) + 31L^4u + 31L^5, \\
B &= 5u^4 + 10u^2\widetilde{D} + \widetilde{D}^2 + 32L^2(3u^2 + \widetilde{D}) + 2L^3u + 31L^4.
\end{aligned}
$$

Thus $A + B\widetilde{d} = \Phi_L(u + \widetilde{d})$ when $\widetilde{d}^2 = \widetilde{D}$. Finally set

$$S = IL^5, \qquad C_a = (2Q - 1)V, \qquad Z = S(n + C_a) - A.$$

With $\mathcal{F}_5$ denoting the ordinary 32-sign norm polynomial, the complete equation is

$$\boxed{\mathcal{P}_{12} = \mathcal{F}_5(Z; B^2\widetilde{D}, 4S^2D_a, 36S^2D_f, 324S^2D_\tau, 2916S^2D_g) = 0.}$$

There is no denominator clearing in this norm: all five radicands are integer polynomials. The displayed homogeneous notation is expanded in the actual DAG. No square root, division, variable exponent or additional quantified register occurs in the exported equation.

1.2 Exclude the entire zero multiplier branch first

For all integer inputs q is nonzero, even and 2 modulo three; $X \equiv 2 \pmod 3$ and $a \in 6\mathbb{Z}$. Hence $\Delta = (a+1)(a+3) \geq 3$, and H is nonzero, odd and divisible by three. Also $H \equiv 3 \pmod q$, so $\gcd(H, M) = 1$ and $L \neq 0$. Thus $D_m \geq 1$ and $\widetilde{D} > 0$, including $v = 0$.

The imported elementary quintic lemma says that π has no zero in a real extension of $\mathbb{Q}$ of degree at most two. The quadratic factor has no real zero, while the cubic has degree three and is irreducible by the rational-root test. Consequently

$$A + B\sqrt{\widetilde{D}} \neq 0, \qquad A - B\sqrt{\widetilde{D}} \neq 0.$$

If $I = 0$, then $S = 0$, the last four norm radicands vanish, and the entire polynomial specializes to

$$(A^2 - B^2\widetilde{D})^{16} \neq 0.$$

Every zero therefore has $I, S, s_0, r, c \neq 0$. This is established before dividing a factor relation by S. Since $Q = (\Delta I)^2$, at such a zero $Q > 1$ and $\delta \neq 0$.

1.3 Small fixed weights replace all size separators

Modulo three the new carrier has

$$M \equiv 2, \quad L \equiv 0, \quad b_0 \equiv 2, \quad u \equiv 2, \quad \widetilde{D} \equiv 1.$$

Thus $B \equiv 5 + 10 + 1 \equiv 1 \pmod 3$. Also

$$D_f \equiv D_\tau \equiv D_a \equiv D_g \equiv 1 \pmod 3.$$

For D_a use $Q \equiv 0 \pmod 9$; for D_g use $s = 3s_0$. For D_τ, the factorization $1 + (XY^2)(XY^2 + 1)k^2$ has $XY^2 \equiv 0$ or 2 modulo three.

Divide a vanishing factor by the already nonzero S. The five weighted roots

$$\frac{BMd}{S}, \quad 2\widetilde{y}, \quad 6f, \quad 18\tau, \quad 54g$$

have squared three-adic orders respectively

$$-2v_3(S), \quad 0, \quad 2, \quad 4, \quad 6,$$

where $d^2 = D_m$, $\widetilde{y}^2 = D_a$, $f^2 = D_f$, $\tau^2 = D_\tau$, $g^2 = D_g$. Each subset of at least two has a unique even minimum. Normalize its smallest root by the corresponding power of three and multiply a putative zero sum over all signs of the other roots. The product is a polynomial in the rational squared roots and is a nonzero unit modulo three, a contradiction. Subtracting a Galois conjugate of the rational factor relation would give just such a forbidden signed subsum. All roots are therefore rational. The nonzero rational multiplier BM/S causes no loss: d is rational too. Roots of integer radicands are integral. No size estimate, radical positivity assumption or bound on R was used for this extraction. Global monotonicity of the chosen quintic, although true, is not needed by this new proof: $B \equiv 1 \pmod 3$ already supplies its nonvanishing.

Now $D_g = g^2 \geq 0$. Since $s \neq 0$, the alternative $P \leq 0$ would give $D_g = (3P - 4)s^2 + 1 \leq -3$. Thus $P > 0$, and the joint-positivity identity gives $d_g > 0$ and $\eta\zeta > 0$. The rational value $y = \widetilde{y}/\delta$ satisfies

$$QV^2 - (Q - 1)y^2 = 1.$$

Adjacent-coefficient Pell descent makes y integral.

1.4 One monic carrier restores both congruences

After absorbing the factor signs into the integer roots, the norm relation gives

$$S \mid A + BMd = \Phi_L(Md - \widetilde{b} + LT).$$

In particular L^5 divides this homogeneous monic quintic. For a prime with $p^a \| L$, if its argument had order $b < a$, the monic fifth-power term would have order $5b$, strictly below every other term. That contradicts divisibility by p^{5a}. Therefore

$$L \mid M(d - b_0) - He.$$

Reduction modulo the coprime factors M, H gives

$$M \mid e, \qquad H \mid d - b_0, \qquad v = q^2(2v_{\text{old}} + 1), \qquad \gamma = (d - b_0)/H \in \mathbb{Z}.$$

Thus $Y = (2v_{\mathrm{old}} + 1)q^3$ is recovered with its full odd quotient. Moreover the quintic argument divided by L is exactly

$$T + \gamma - e/M.$$

This proves both congruences without adding a variable or presupposing that v is nonzero.

The late source's direct normalized signed-kernel proof now applies. The physical coordinates satisfy its first, main and normalized strong Pell equations, strict guards and native transport/index equations. Its two auxiliary quotients are restored by

$$o = f(c - R + 4IT),$$
$$j_{\mathrm{aux}} = \Delta s^2 r^2 c^3 (c - R) + 4srcD_f T,$$

so $V = fo - c = cj_{\mathrm{aux}} - R$. The normalized rank argument uses $c^2 \mid I$ and does not require I/c^2 to be odd. Accordingly, we do not assert that the old coordinate $(\Delta sr - 1)/2$ is integral. The inherited signed step-down, residue orientation and native compiler then produce a finite syntactic refutation, with their ordinary theorem status unchanged.

1.5 Completeness and the exceptional guard value

Start with the late source's canonical native witnesses and set $v = q^2(2v_{\mathrm{old}} + 1)$. Choose its nearest-square coordinate t so $P > 0$. A crucial exception must be excluded before forcing $3 \mid s$: if $P = 1$, the guard equation would be $g^2 + s^2 = 1$ and no nonzero multiple of three could serve as s.

Here that exception is impossible. Both cone products and q are even, so $d_g \equiv 0 \pmod 4$. Therefore

$$P \equiv -(t^2 - d_g - \eta\zeta + 1)^2 \equiv 0 \text{ or } 3 \pmod 4.$$

Positive P is at least three. Thus $D = 3P - 4$ is positive and 2 modulo three, hence nonsquare. Take a positive Pell pair $g_1^2 - Ds_1^2 = 1$. Modulo three, $g_1^2 + s_1^2 = 1$, so $3 \mid g_1 s_1$. Squaring the unit gives

$$g = g_1^2 + Ds_1^2, \qquad s = 2g_1 s_1 > 0, \qquad 3 \mid s.$$

Set $s_0 = s/3$. No outer coordinate has changed.

Invoke the imported full-residue interpolation lemma with $N = s$. It supplies a single normalized Pell pair I, f, with $4sc^2 \mid I$, *before* a residue is requested, and then realizes every class of T modulo this same I in

$$V = f^2(c - R + 4IT) - c, \qquad (\Delta I)^2 V^2 - ((\Delta I)^2 - 1)y^2 = 1.$$

Put $r = I/(sc^2)$. The lemma's hypotheses include even $a + 2$, $R \equiv c \equiv 3 \pmod 4$, all supplied by the canonical source. Its proof uses the corrected period $4n_0 f$ modulo f^2 and the Lucas valuation bound $\gcd(4n_0 f, 4I^2) \mid 4I$; mere infinitude of Pell solutions would not suffice.

Choose an intersective residue for $\pi(T + \gamma - e/M)$ modulo I. Interpolation supplies T, y in that class, leaving I, f fixed. Hence

$$A + BMd = L^5 \pi(T + \gamma - e/M)$$

is divisible by $S = IL^5$. With $\widetilde{y} = \delta y$, choose

$$n = \frac{A + BMd}{S} + 2\widetilde{y} + 6f + 18\tau + 54g - (2Q - 1)V.$$

This is integral and makes one norm factor vanish. The witness order is native data, then t, s, s_0, then I, f, r, then the intersective residue and T, y, and finally n. There is no circular residue choice.

1.6 Exact degree 2048

The relevant polynomial degrees are

register	H	Δ	P	I	D_f	Q	V	D_a	D_g	L
degree	4	8	28	4	16	24	21	90	30	6

and

$$\deg A = 35, \quad \deg B = 28, \quad \deg S = 34, \quad \deg C_a = 45, \quad \deg D_\tau = 22, \quad \deg \widetilde{D} = 14.$$

Pair the two auxiliary signs first. Let B_* denote $Sn - A$ plus the four signed non-auxiliary radical terms. Their weights are $35, 42, 45, 49$ for main, strong, first and guard, so B_* has weight at most 49. The paired factor is exactly

$$B_*^2 + 2SC_aB_* + S^2(V^2 + 4Q - 4),$$

because the polynomial identity

$$((2Q-1)V)^2 - 4(Q-1)(QV^2-1) = V^2 + 4Q - 4$$

holds universally. Its term bounds are $98, 128, 110$. Taking sixteen sign choices proves degree at most $16 \cdot 128 = 2048$. The unique leading paired term is $2SC_a(\pm 54S\sqrt{D_g})$; the eight negative signs cancel. Thus the full leading form is

$$108^{16} S_0^{32} (C_a)_0^{16} (D_g)_0^8 \neq 0.$$

More explicitly, the coefficient of

$$j^{928} w^{288} v^{320} h^{32} s_0^{128} r^{112} c^{224} T^{16}$$

is

$$\boxed{2^{2064} 3^{1400} \beta^{928} > 0.}$$

The exponent sum is 2048. The h^{32} term uniquely selects ha_0 throughout the top part of k^{32}; T^{16} then uniquely selects $4I_0T$ throughout $(4I_0T - R_0)^{16}$. The remaining factors are monomials, so no other fixed coefficient can cancel this term.

Independent actual-coefficient line certificates modulo 65521 and 65519 also attain degree 2048. The complete 233-operation DAG has no dead gates or hidden input leaves. Independent arithmetic checks use a separate binomial quintic remainder and full 32-sign product, including every intersection of the exceptional slices $s_0 = 0, r = 0, c = 0$. These finite checks do not replace the unbounded interpolation or native calibration proofs. The resulting consistency equivalence is an ordinary mathematical theorem composed with those explicitly retained sources, not a proof of consistency or a completed object-language ZFC derivation.